

## 1 Semi-Implicit

$$\begin{aligned}
\frac{1}{\Delta t}((h\mathbf{u})' - h^{(n)}\mathbf{u}^{(n)}) &= -\nabla \cdot \left( h^{(n)}\mathbf{u}^{(n)}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{i-1}\alpha\mathbf{u}^{i-1} \right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
&\quad - (h^{(n)}(1-\alpha) + h^{i-1}\alpha)g\nabla h_0 \\
\frac{1}{\Delta t}(h^i - h^{(n)}) &= -\nabla \cdot \left( (1-\alpha)h^{(n)}\mathbf{u}^{(n)} + \alpha(h\mathbf{u})' \right) \\
&\quad + \alpha^2(\Delta t)h^{i-1}g\nabla^2(h^i) \\
\frac{1}{\Delta t}(h^i\mathbf{u}^i - (h\mathbf{u})') &= -h^{i-1}\alpha g\nabla(h^i)
\end{aligned}$$

## 2 Segregated Implicit Advection, Implicit Gravity Waves

$$\begin{aligned}
\frac{1}{\Delta t}(h^{i-1}\mathbf{u}' - h^{(n)}\mathbf{u}^{(n)}) &= -\nabla \cdot \left( h^{(n)}\mathbf{u}^{(n)}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{i-1}\alpha\mathbf{u}' \right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
&\quad - h^{(n)}g\nabla(1-\alpha)(h^{(n)} + h_0) - h^{i-1}g\nabla\alpha(h^{i-1} + h_0) \\
\frac{1}{\Delta t}(h^{i-1}\mathbf{u}'' - h^{(n)}\mathbf{u}^{(n)}) &= -\nabla \cdot \left( h^{(n)}\mathbf{u}^{(n)}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{i-1}\alpha\mathbf{u}' \right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
&\quad - h^{(n)}g\nabla(1-\alpha)(h^{(n)} + h_0) \\
\frac{1}{\Delta t}(h^i - h^{(n)}) &= -\nabla \cdot (h^{i-1}\mathbf{u}'') + \alpha^2(\Delta t)h^{i-1}g\nabla^2(h^i + h_0) \\
\frac{1}{\Delta t}(\mathbf{u}^i - \mathbf{u}'') &= -\alpha g\nabla(h^i + h_0)
\end{aligned}$$

## 3 Operator Split Implicit Advection, Implicit Gravity Waves

$$\begin{aligned}
\frac{1}{\Delta t}(h^{i-1}\mathbf{u}' - h^{(n)}\mathbf{u}^{(n)}) &= -\nabla \cdot \left( h^{(n)}\mathbf{u}^{(n)}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{i-1}\alpha\mathbf{u}' \right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
\frac{1}{\Delta t}(h^i - h^{(n)}) &= -\nabla \cdot \left( (1-\alpha)h^{(n)}\mathbf{u}^{(n)} + \alpha h^i\mathbf{u}' \right) \\
&\quad + \alpha(\Delta t)h^{(n)}g\nabla^2(1-\alpha)(h^{(n)} + h_0) - \alpha(\Delta t)h^{i-1}g\nabla^2\alpha(h^i + h_0) \\
\frac{h^i}{\Delta t}(\mathbf{u}^i - \mathbf{u}') &= -h^{(n)}(1-\alpha)g\nabla(h^{(n)} + h_0) - h^i\alpha g\nabla(h^i + h_0)
\end{aligned}$$